ChatGPT

Point-in-Time Regime Framework for Crypto Perpetual Futures

Design logic

A practical regime layer for crypto perpetual futures should be hierarchical rather than monolithic. The literature does support trend and momentum in crypto, but not as a stable “always on” anomaly. Cross-sectional crypto returns are strongly shaped by a market factor and a momentum factor; a separate trend factor built from price and volume also predicts the cross-section robustly across subsamples; and lagged returns of other cryptocurrencies predict focal-coin returns out of sample on Binance data. At the same time, crypto momentum is unstable, sample-sensitive, and crash-prone, especially when it is implemented indiscriminately without risk management or liquidity filters. That combination argues for a regime layer whose job is not to forecast exact returns, but to decide when breakout and continuation families should be allowed to express themselves, when they should trade smaller, and when they should be turned off. [1]

The second reason to use a layered regime framework is that crypto perpetuals are not just price series. Funding, open interest, mark-index deviations, and microstructure conditions affect whether a move is clean, crowded, or vulnerable to forced unwinds. Crypto microstructure measures of liquidity and price discovery have predictive power for price dynamics; futures often lead spot price discovery, with leadership increasing around macro surprises; spot market quality around perpetuals follows a U-shape over the eight-hour funding cycle; and standardized carry predicts subsequent short-side liquidations, which is direct evidence that derivatives crowding matters for continuation and crash risk. Intraday volatility and volume also show systematic daily, weekly, and within-hour periodicity that is plausibly linked to algorithmic trading and funding times. [2]

The framework therefore should have four layers: an eligibility layer that enforces tradability and point-in-time universe integrity; a market-weather layer built from BTC, ETH, breadth, and alt-risk appetite; a derivatives overlay that labels clean trend, crowded positioning, and deleveraging; and a timing layer for session windows and catalysts. The hard gates should live mostly in the eligibility and liquidity layers, with a few extreme derivatives states added. Market-weather and session variables should mostly modulate sizing and strategy permissions rather than attempt to classify the whole market with a single opaque latent-state model. That structure matches the evidence: crypto trends exist, but their payoffs depend heavily on crowding, liquidity, and the composition of the active universe. [3]

Layer	Purpose	What it controls
Eligibility and market hygiene	Ensure the contract is tradeable, live, and not structurally broken	Universe membership, hard disable
Market weather	Decide whether the market favors continuation or defense	Strategy activation and baseline size

Layer	Purpose	What it controls
Derivatives overlay	Detect crowding, squeeze risk, and resets	Size cuts, temporary disables, fade permission
Session and catalyst overlay	Improve timing around funding, opens, listings, and event windows	Entry timing, slippage assumptions, catalyst-only strategies

This architecture is a synthesis of the evidence above and of exchange-native data availability from Binance, Bybit, and OKX, all of which expose the ingredients needed for point-in-time construction: mark price, index price, premium/funding inputs, open interest, instrument metadata, and delisting status. [4]

Regime feature dictionary

The right question is not "does this variable predict returns?" but "does this variable improve the decision of whether a strategy family should be active here?" The table below separates variables into high-confidence, medium-confidence, and low-confidence groups based on three things: direct evidence in crypto studies, clean point-in-time availability from exchange data, and susceptibility to overfit or taxonomy drift. The evidence judgments are a synthesis of work on crypto factor structure, cross-crypto predictability, momentum instability, funding and carry, exchange listings, delisting bias, price discovery, fragmentation, and microstructure. [5]

Feature family	Concrete features	Evidence in crypto	PIT availability	Overfit / noise risk	Recommended use
Parent market trend	BTC and ETH above 20/50/100-day MAs; normalized slope of 20/50-day MA; 20/60/90-day drawdown; short-term vs medium-term alignment	High	High	Low	Core gate for momentum and breakout longs
Market breadth	Percent of active universe above 20/50/100-day MAs; percent at 20/60/90-day highs; advance/decline ratio; cross-sectional median return; dispersion of 1D and 5D returns	High to medium	High, if universe is PIT-clean	Medium	Confirm or veto breakouts; distinguish broad continuation from leader-only noise

Feature family	Concrete features	Evidence in crypto	PIT availability	Overfit / noise risk	Recommended use
Alt-risk appetite	ETHBTC; SOLBTC; equal-weight liquid alts vs BTC; equal-weight liquid alts vs ETH; liquid alt basket relative strength; BTC-perp turnover share or OI share as exchange-native BTC-dominance proxy	Medium	High for ratios, medium for dominance proxies	Medium	Sizing modifier; activation filter for sector continuation and prior-high momentum
Derivatives state	Price × OI matrix; OI change over 8h/24h/3d; funding percentile; funding sign persistence; mark-index premium percentile; premium-index stress; crowded-long / crowded-short / reset labels	High for funding, stress, carry; medium for raw OI matrix	High	Medium	Overlay for crowding, squeezes, deleveraging resets, risk cuts
Liquidity state	Dollar volume; turnover; best-bid/ask spread or spread proxy; wick/range proxy; stale-bar ratio; zero-volume bars; venue anomaly flags	High	High	Low	Hard gate for all strategies
Session state	Distance from UTC daily open; distance from 4h turns; distance from funding timestamps; US cash open windows; weekend/weekday flag	Medium	High	Medium to high	Timing and sizing only, not hard gate except around severe stress
Lifecycle and catalyst	Listing age; launch timestamp; delist schedule or notice; exchange announcements; event window flag; sector-ignition breadth from a fixed sector map	High for listing/ delisting; medium for sector ignition	High for exchange-native items	Medium	Hard gate for new or delisting contracts; activation key for post-catalyst continuation

Feature family	Concrete features	Evidence in crypto	PIT availability	Overfit / noise risk	Recommended use
External optional	Raw BTC dominance from broad market-cap providers; narrative/ sentiment labels; dynamic sector taxonomies from third parties	Low to medium	Medium to low	High	Optional research variables only, not core production gates

The most robust variables are the ones that sit closest to three established facts. First, crypto has a strong common market component, so parent BTC/ETH trend and cross-sectional breadth are not redundant; they answer different questions. Second, continuation strategies are fragile when returns are concentrated in a small subset of names or depend on extreme tail events, so breadth and dispersion matter as quality controls. Third, derivatives crowding and liquidity conditions change the payoff distribution of momentum more than they change the sign of the historical chart pattern. That is why parent trend, breadth, liquidity, and lifecycle belong in the core grammar, while session labels, raw dominance, and narrative-sector labels should stay secondary. [6]

A few variables should be treated explicitly as “useful but not standalone.” The price-plus-open-interest matrix is one of them. Open interest is definitionally the stock of open contracts, not a directional signal by itself, so its main value is structural: rising price with rising OI usually says new participation is building, while large OI drops after a sharp move look more like forced closure or a reset. In crypto perps, that interpretation becomes much stronger when it is combined with funding percentile and mark-index stress, because carry and funding extremes are tied to liquidation risk. Use price × OI as a diagnostic overlay, not as a primary hard gate on its own. [7]

On BTC dominance, the safest answer is that the classical metric is not exchange-native. Dominance is defined as Bitcoin market cap divided by total crypto market cap, which requires an external market-cap universe whose membership changes through time. If the mandate is “computed mostly from exchange data,” replace raw BTC dominance with an exchange-native proxy such as BTC-perp share of top-universe turnover or OI, or the relative-strength spread between a liquid-alt equal-weight basket and BTC. Those proxies are uglier, but they avoid a hidden external universe dependency. [8]

Discrete labels and activation map

The cleanest way to label regimes without leaking future information is to keep the labels discrete, interpretable, and layered. I would not fit a single all-in HMM or clustering model as the production regime classifier. In crypto, connectedness, spillovers, and the profitability of continuation signals move around materially across episodes, and momentum payoffs can be driven by rare tails. A coarse label grammar that is fixed ex ante and updated only through rolling percentiles is easier to audit, easier to stress, and less likely to bend around one cycle. [9]

Use the following discrete labels:

Layer	Label	Working definition
Parent market	**Expansion**	BTC and ETH above medium-term trend, slopes positive, drawdowns contained, breadth strong
Parent market	**Fragile uptrend**	BTC and/or ETH still above medium-term trend, but breadth deteriorating, alt appetite fading, or derivatives crowding elevated
Parent market	**Rotation / mixed**	Parent trend mixed, breadth mixed, but enough participation and relative-strength dispersion to support selective leaders
Parent market	**Countertrend bounce**	Parent market below slow trend but above short MA after a selloff; breadth still mediocre; rallies prone to fail
Parent market	**Stress**	BTC and ETH below medium-term trend, slopes negative, deeper drawdowns, broad weak participation
Derivatives overlay	**Clean**	Funding near neutral percentiles, premium stress low, OI behavior consistent with orderly trend
Derivatives overlay	**Crowded long**	Funding in upper tail, premium positive and persistent, OI rich, price not advancing proportionately
Derivatives overlay	**Crowded short**	Funding in lower tail, negative premium persistence, OI rich, price not falling proportionately
Derivatives overlay	**Reset / deleveraging**	Sharp OI contraction with funding normalization and reduced premium stress after a squeeze or flush
Liquidity overlay	**Normal**	Volume, spread, and stale-bar measures in normal percentiles
Liquidity overlay	**Thin**	Tradable, but wider spreads, lower turnover, higher wickiness, weaker execution quality
Liquidity overlay	**Stressed**	Venue anomalies, stale bars, spread spikes, extreme bar pathologies
Lifecycle overlay	**Mature**	Sufficient price history and no known delist risk
Lifecycle overlay	**New listing discovery**	Very young contract, limited history, unstable behavior
Lifecycle overlay	**Catalyst active**	Time-stamped exchange event window or sector ignition present
Lifecycle overlay	**Delist risk**	Delist schedule present, contract settlement imminent, or venue notice active

The first table below shows the parent-regime activation map. “Active” means the strategy can run at its baseline sizing before overlays. “Reduced” means allowed, but only after the derivatives and liquidity overlays cut size. “Disabled” means the strategy family is off except for already-open positions being managed out. This is the part of the framework that should stay stable over time. The logic follows the evidence that trend-following and cross-sectional continuation work best in liquid, tradable subsets and can be undermined by crashes, broad stress, and tail-driven crowding. [10]

Strategy family	Expansion	Fragile uptrend	Rotation / mixed	Countertrend bounce	Stress
Liquid leader breakout long	Active	Reduced	Reduced	Disabled	Disabled
Prior-high momentum	Active	Reduced	Active for top leaders only	Disabled	Disabled
Time-series momentum	Active	Active at lower size	Reduced / selective	Selective only	Active only if mandate allows short-side trend; otherwise disabled
Post-breakout retest continuation	Active	Reduced	Reduced / selective	Disabled	Disabled
Sector leader continuation	Active when sector ignition present	Reduced	Active when alt appetite positive and sector breadth rising	Disabled	Disabled
Post-catalyst continuation base	Active only with catalyst	Active only with catalyst	Active only with catalyst	Selective only, with tighter stops and smaller size	Disabled except special event books
Risk-off de-risk overlay	Low intensity	Medium intensity	Medium intensity	High intensity	Maximum intensity
Failed breakout / fade	Disabled	Selective only	Selective only	Active as secondary	Active as secondary

The second table is the overlay map. These overlays should dominate the parent label whenever execution quality or crowding gets extreme.

Overlay condition	Liquid breakout and momentum longs	Sector continuation	Post-catalyst continuation	Failed breakout / fade	Risk-off overlay
Clean derivatives + normal liquidity	Keep baseline	Keep baseline	Keep baseline if catalyst active	Usually off	Low
Crowded long	Cut size materially; disable if breadth is weak too	Cut size materially	Allow only if catalyst is very fresh and liquidity strong	Permission increases	Raise
Crowded short	Little change for longs; watch squeeze risk	No change to mild increase	Mild increase if catalyst aligns	Fade shorts cautiously	Mild
Reset / deleveraging after flush	Re-enable leaders, but start at half size	Re-enable leading sector	Good environment for post-catalyst continuation if premium stress has normalized	Reduce	Ease gradually
Thin liquidity	Cut size or disable	Cut sharply	Only trade if catalyst edge is large	Reduce frequency	Raise
Stressed liquidity or venue anomaly	Disable new entries	Disable new entries	Disable new entries	Disable unless specialist strategy book	Maximum
New listing discovery	Only catalyst-specific book, tiny size	Off	Active if this is the catalyst event itself	Off	Raised
Delist risk	Disable	Disable	Disable	Disable	Maximum

The main principle is simple. Trend and continuation families should require both directional permission and market-quality permission. The parent market label tells you whether continuation is allowed in principle. Breadth and alt appetite tell you whether participation is wide enough. Derivatives overlays tell you whether the move is orderly or crowded. Liquidity and lifecycle decide whether you should trade it at all. Failed-breakout logic should remain subordinate because crypto continuation can persist longer than conventional mean-reversion priors would suggest when the market and derivatives tape are aligned. [11]

Point-in-time construction rules, data requirements, and leakage traps

Point-in-time construction matters at least as much as feature choice. Binance and Bybit both expose mark price, index price, premium or premium-index series, funding history, and instrument metadata; Bybit documents that funding is updated through a minute-level process tied to interest rate and premium index; and Binance exposes delist schedules and symbol exchange information. Bybit also documents that mark price is derived from a spot index plus a decaying funding-basis component, and that mark price is the liquidation trigger. These details matter because your regime layer should be aligned to the exchange's actual liquidation and funding mechanics, not to an end-of-day chart abstraction. [12]

The core point-in-time rules should be written down as production invariants:

Rule	Practical requirement
Universe membership	At time *t*, include only contracts whose status is live and tradable at *t*, using exchange instrument metadata and launch timestamps
Minimum history	Exclude contracts from general momentum and breakout books until they have enough lookback to compute all required features; allow them only in the new-listing catalyst sleeve
Breadth universe	Compute breadth on the active tradable universe at *t*, not on today's surviving universe
Delist handling	Keep a contract in history until its actual delist time; after the notice appears, move it into delist-risk state immediately
Relative strength consistency	Compute ETHBTC, SOLBTC, and basket-vs-BTC ratios from a consistent venue or from synchronized index-price composites
Percentiles	Funding, premium stress, liquidity, and dispersion thresholds must be rolling or expanding up to *t* only; never use full-sample percentiles
OI alignment	Use OI snapshots time-aligned to the decision bar and, if publication timing is uncertain, lag by one bar
Session flags	UTC open, 4h turns, funding windows, and weekend flags are deterministic and PIT-safe
Catalyst labels	Use only time-stamped exchange-native events or a separately versioned event feed with recorded publication timestamps
Sector labels	Fix sector mappings in a versioned file and change them only on scheduled rebalance dates

These rules are directly motivated by the empirical problems the crypto literature keeps surfacing: delisting bias is large; listings and venue events materially move prices; price efficiency depends on market structure; and wash trading can contaminate reported activity. A live universe and a dead-sure event clock are not housekeeping details here. They are part of the alpha-defense system. [13]

For data, I would separate the stack into minimal, preferred, and optional inputs.

Tier	Required inputs
Minimal	Contract OHLCV; mark price; index price; funding history; open interest history; instrument metadata including launch time and status; delist schedule or notice history
Preferred	Top-of-book bid/ask; premium index history; trade prints; order-book depth snapshots; exchange announcement archive for listings, delistings, maintenance, and contract changes
Optional	Liquidation feeds; top-trader ratios; cross-venue basis; external market-cap dominance; external sector taxonomies

That minimal tier is enough to build the framework in this report. The preferred tier materially improves execution realism and spread proxies, and it lets you replace crude wick-based or range-based liquidity proxies with direct bid/ask and depth measures. The optional tier is research-only until you have verified timestamp quality and venue coverage. [14]

The most important leakage traps are specific to crypto datasets rather than generic to time series:

Leakage trap	Why it is dangerous
Survivorship and delisting bias	Keeps only winners; can massively inflate equal-weight or small-cap results
Universe creep	Using today's active contracts to compute old breadth or old leader ranks silently leaks survival
Full-sample normalization	Percentiles or z-scores computed on the whole sample leak future stress regimes backward
External dominance without historical snapshots	Raw BTC dominance needs a changing market-cap universe and can import hidden lookahead
Event timestamp slippage	Scrape time is not publication time; announcement pages can be edited later
OI and funding misalignment	If the backtest assumes publication earlier than reality, the derivatives overlay sees the future
Using 24h ticker fields as if they were bar-local	Rolling exchange ticker statistics can be misaligned with decision bars
Ignoring contract spec changes	Tick size, lot size, leverage rules, and margin logic can change and affect fill realism
Taxonomy revisionism	Retagging sectors with hindsight can create fake "sector ignition" history

Leakage trap	Why it is dangerous
Cross-venue stitching without synchronization	Apparent basis or divergence can be a timestamp problem rather than a regime feature

If the team follows only one meta-rule, it should be this: every feature must be reproducible from a historical snapshot that you could have had at the decision timestamp, using the universe and metadata that existed then. The exchange docs provide the mechanics; the studies on survivorship, listings, fragmentation, and price discovery explain why shortcuts are expensive. [15]

Validation method and parameter ranges

Strategy-regime interactions should be validated in a nested process, not by fitting the regime grammar on the whole sample and then reporting conditional Sharpe ratios. Standard holdout methods are weak in trading research because they understate path dependence and multiple testing. The right toolkit here is rolling walk-forward for deployment realism, purged and embargoed cross-validation to avoid overlap leakage, CPCV when you want multiple plausible test paths, and post-selection controls such as the Deflated Sharpe Ratio, Probability of Backtest Overfitting, SPA, and Model Confidence Set. Those tools exist for exactly the situation you are in: many related strategies, many plausible regime filters, and a strong temptation to crown the best conditional backtest after a large specification search. [16]

A workable validation pipeline is:

Component	Recommendation
Outer evaluation	Rolling walk-forward with chronological retraining only
Training window	18 to 36 months for daily features; longer if using many percentile features
Test window	1 to 3 months for research; 1 month is closest to live monitoring cadence
Inner model selection	CPCV or purged k-fold inside the training window only
Purging	Remove training samples whose label horizon overlaps the test window
Embargo	At least the maximum holding period, and never less than one funding interval for perp strategies
Objective	Choose regime rules by OOS utility of the strategy family, not by isolated feature prediction accuracy
Multiple testing control	Report DSR and PBO for every family; use SPA or White-style reality-check logic for competing gating rules; use MCS for model sets that remain statistically tied
Rare regimes	Collapse or merge labels that do not achieve minimum frequency and episode diversity
Stability test	Require regime advantage to persist across market episodes, not just across folds

On regime frequency, the right rule is conservative. A label should not enter production unless it appears often enough to estimate both mean and tail behavior. As a practical minimum, require each production regime to occur in at least three distinct macro episodes and to generate roughly thirty or more independent strategy events in the combined out-of-sample test paths. If not, merge it upward into a broader parent label. This is not a theorem; it is an anti-fragility rule meant to stop the team from operationalizing a label that is really a single historical anecdote. That recommendation is consistent with the literature's broader warning that uninformative data should leave you with broader confidence sets, not finer model claims. [17]

The parameter ranges below are not "optimal settings." They are coarse starting bands designed to reduce specification search. They are grounded in the broad evidence for trend and microstructure effects in crypto, the instability of raw momentum, and the practical need to use low-dimensional, interpretable filters. [18]

Component	Recommended starting range
Slow trend windows	80 to 130 trading days
Medium trend windows	40 to 70 trading days
Fast trend windows	15 to 30 trading days
MA slope window	10 to 20 trading days, normalized by ATR or realized vol
Parent drawdown window	20 to 90 trading days
Breadth thresholds	Weak below roughly 40% above MA50, neutral around 40% to 60%, strong above roughly 60%
New-high breadth windows	20, 60, and 90 trading days
Dispersion window	20 trading days for daily strategies; 5 trading days as a short-horizon companion
Funding percentile window	90 to 365 calendar days, computed per symbol or per venue-symbol group
Funding persistence	2 to 6 funding intervals of same sign
OI change windows	8h, 24h, and 3d
Premium or mark-index stress	Upper or lower 5% to 10% of rolling history
Liquidity floor	Top half to top third of universe by rolling median dollar volume
Stale-bar disable threshold	If more than roughly 1% to 3% of recent bars are stale or zero-volume, disable
New-listing general-strategy age filter	20 to 60 days before entry into non-catalyst books
Catalyst continuation window	Event day through roughly 1 to 10 trading days after the event, with faster decay for listings

The practical split between hard gates and sizing modifiers should also stay coarse. Hard gates should include: stressed-liquidity state, venue anomaly state, delist-risk state, insufficient listing age for general books, and severe breadth failure for long-only continuation families. Sizing modifiers should include: strength of breadth, alt-risk appetite, funding percentile, derivatives crowding, weekend or funding-window timing, and relative liquidity rank. Treating too many variables as binary gates is a classic way to turn a risk-control system into a disguised optimizer. [19]

Minimal implementation plan and backtesting-agent prompt

A minimal backtesting implementation does not need a giant feature store on day one. It needs a clean instrument master, a point-in-time active universe, deterministic feature transforms, a layered label grammar, and a way to attribute PnL changes to regime decisions rather than strategy alpha. The first production question to answer is not "does the regime layer improve Sharpe?" but "does it improve conditional deployment discipline without hiding new forms of leakage?" That means you should store every daily regime snapshot, every reason code for disable or size cut, and every strategy decision before running any optimization loops. [20]

The implementation sequence should look like this:

Phase	Deliverable
Instrument master	Versioned contract metadata with launch time, status, tick/lot specs, exchange, sector tag, and delist notices
Data build	Daily and 4h aligned bars for trade price, mark price, index price, funding, OI, and volume
Active universe	Point-in-time contract set with history-age, liquidity, stale-bar, and delist filters
Feature engine	Parent trend, breadth, alt-risk appetite, derivatives overlays, liquidity overlays, session flags, lifecycle flags
Label engine	Parent market label plus derivatives, liquidity, and catalyst overlays
Activation engine	Strategy permission matrix returning active / reduced / disabled state and target size multiplier
Backtest harness	Strategy family backtests run both unconditional and regime-conditioned with funding and execution costs
Validation layer	Rolling walk-forward, inner CPCV or purged CV, DSR, PBO, SPA or MCS, and regime-frequency diagnostics
Monitoring	Daily snapshot report of label counts, universe changes, size reductions, and disagreement between layer signals

The prompt below is designed for an internal backtesting agent. It assumes Python, daily plus 4h data, and a requirement to keep every transformation point-in-time safe.

```
You are implementing a point-in-time regime/context layer for crypto perpetual
futures.

Objective:
Build a deterministic, auditable regime engine that decides whether each
strategy family is ACTIVE, REDUCED, or DISABLED on each bar. The regime layer
must be computed mostly from exchange data and must avoid lookahead,
survivorship bias, and post-hoc relabeling.

Data inputs:
- Instrument metadata by timestamp: symbol, exchange, contract type, launchTime/
onboardDate, status, tick size, lot size, settlement/delist info.
- Bars by symbol and timestamp: trade close/high/low/open/volume, mark price,
index price, funding rate, open interest.
- Optional: best bid/ask, premium index, order book snapshots, exchange
announcements, sector tags.
- Timezone standard: UTC for all bars and event times.

Core outputs per symbol and date:
- active_universe_flag
- liquidity_state
- lifecycle_state
- parent_market_state
- breadth_state
- alt_risk_state
- derivatives_state
- session_flags
- catalyst_flags
- final_activation_map by strategy family
- size_multiplier by strategy family
- reason_codes explaining every disable or size cut

Mandatory implementation rules:
- Use point-in-time universe membership only.
- Never compute breadth on today’s surviving universe.
- Never normalize with full-sample statistics.
- Percentiles must be rolling or expanding with history available up to t only.
- Delisting notices must affect the state from their publication/effective time
forward.
- New listings must remain out of general momentum/breakout books until minimum
history is satisfied.
- If OI/funding timestamp alignment is uncertain, lag those features by one bar.
- Persist all intermediate states to disk for audit.

Feature construction:
- Parent trend:
  - BTC and ETH above MA20, MA50, MA100
```

```
  - normalized slopes for MA20 and MA50
  - rolling drawdowns over 20d, 60d, 90d
- Breadth on active universe:
  - pct above MA20/50/100
  - pct at 20d/60d/90d highs
  - advance/decline ratio
  - cross-sectional 1d and 5d return dispersion
- Alt-risk appetite:
  - ETHBTC return and trend
  - SOLBTC return and trend
  - equal-weight liquid-alt basket vs BTC
  - equal-weight liquid-alt basket vs ETH
  - exchange-native BTC dominance proxy = BTC share of rolling universe turnover
or OI
- Derivatives:
  - rolling changes in OI over 8h, 24h, 3d
  - rolling funding percentiles per symbol
  - funding sign persistence
  - premium stress using mark-index spread or premium index percentile
  - crowded_long, crowded_short, reset labels from joint price/OI/funding/
premium behavior
- Liquidity:
  - rolling median dollar volume
  - spread proxy from bid/ask if present, else a bar-based proxy
  - wick/range proxy
  - stale-bar ratio and zero-volume counts
  - venue stress flag if bar pathologies exceed threshold
- Session/order-flow:
  - weekend flag
  - minutes/bars from UTC daily open
  - minutes/bars from 4h turns
  - minutes/bars from funding windows
  - US cash open window flag
- Lifecycle/catalyst:
  - listing age
  - delist-risk flag
  - catalyst flag from exchange-native event feed
  - sector ignition flag when a fixed sector basket shows synchronized breadth
expansion

Discrete labels:
- parent_market_state in {EXPANSION, FRAGILE_UPTREND, ROTATION,
COUNTERTREND_BOUNCE, STRESS}
- derivatives_state in {CLEAN, CROWDED_LONG, CROWDED_SHORT, RESET}
- liquidity_state in {NORMAL, THIN, STRESSED}
- lifecycle_state in {MATURE, NEW_LISTING_DISCOVERY, CATALYST_ACTIVE,
DELIST_RISK}
```

```
Strategy families:
- liquid_leader_breakout_long
- prior_high_momentum
- time_series_momentum
- post_breakout_retest_continuation
- sector_leader_continuation
- post_catalyst_continuation_base
- risk_off_overlay
- failed_breakout_fade_secondary

Activation logic:
- Start from parent_market_state.
- Apply liquidity and lifecycle hard gates first.
- Apply derivatives overlays next.
- Apply session/catalyst modifiers last.
- Return ACTIVE / REDUCED / DISABLED plus a numeric size multiplier.
- Failed breakout/fade must remain secondary and should never dominate in clean
expansion states.

Validation requirements:
- Backtest every strategy family both unconditional and regime-conditioned.
- Use rolling walk-forward.
- Use purging and embargo wherever labels or holdings overlap.
- Add an inner CPCV or purged-CV selection loop for threshold tuning.
- Report out-of-sample Sharpe, Sortino, turnover, exposure, hit rate, max
drawdown, tail metrics, and DSR/PBO diagnostics.
- Report conditional performance by regime label.
- Report regime frequency and merge labels that are too rare.

Coding requirements:
- Python only.
- Write production-style code with type hints, comments, and unit-testable pure
functions.
- Keep configuration externalized in YAML or JSON.
- Make all thresholds easy to sweep over coarse grids.
- Log every reason code for disable/reduce decisions.
- Save intermediate feature tables and final daily regime snapshots.

Deliverables:
- feature_engine.py
- label_engine.py
- activation_engine.py
- config.yaml
- tests for PIT safety and universe construction
- notebook or script demonstrating one full walk-forward run
- summary tables comparing unconditional vs regime-conditioned results
```

The two unresolved tensions are the ones that usually matter most in production. First, the more expressive the regime grammar becomes, the closer it gets to becoming a hidden optimizer. Second, the more exchange-native and point-in-time pure the framework becomes, the more it will under-represent non-exchange catalysts that clearly move crypto. The right compromise is to keep the core weather layer exchange-native and hard to fool, then add a small, timestamped catalyst sleeve around it rather than letting the whole system drift into narrative classification. [21]

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